feb. 5

- Review from before
  - Order of a group → cardinality
  - Order of an element → # of copies to get to identity
  - Multiplicative notation for group operations → $h*k = hk$
    - Exponent rules for group operation → $\underbrace{h*h*\ldots*h}_{7 \text{ times}} = \underbrace{h\cdot h\cdot\ldots\cdot h}_{7 \text{ times}} = h^7$

      $h^7\cdot h^{-2} = \underbrace{h\cdot\ldots\cdot h}_{7 \text{ times}}\cdot\underbrace{h^{-1}\cdot h^{-1}}_{2 \text{ times}} = h^{7-2} = h^5$

  - Subgroups → subset of a group that is a group under the same operation

    ex. $\mathbb{Z} < \mathbb{Q}$ (under addition, default operation for $\mathbb{Z}, \mathbb{Q}$ when not specified)

- <u>Thm. 3.1</u> (One-Step Subgroup Test) Let G be a group and H a <u>nonempty</u> subset. Then H is a subgroup if and only if whenever $a, b \in H$, then $ab^{-1} \in H$ ← note this is $a*b^{-1}$, not $(ab)^{-1}$

  If G is a group and $\emptyset \neq H \subset G$, then $H \leq G \Leftrightarrow ab^{-1} \in H$ whenever $a, b \in H$

  <u>Proof</u>

  Consider $(H \neq \emptyset, \cdot)$ where $\cdot$ is the binary operation for G. Suppose ab^{-1} is in H whenever $a, b \in H$. We want to show $H \leq G$.

  Identity: first, $H \neq \emptyset$ so we have elements in H. Thus we can take $x \in H$. By our hypothesis, $xx^{-1} = e \in H$. Hypothesis with $a = x, b = x$

  Associativity: To see H is associative, we note G is associative. Let $a, b, c \in H$. We have $a, b, c \in G$ since $H \leq G$. Thus $(ab)c = a(bc)$ because G is a group & hence associative. We need to make sure this equality makes sense in H, i.e all parts are in H ☆

  Inverses: Let $x \in H$. By hypothesis $ex^{-1} = x^{-1} \in H$. Thus if $x \in H$, then $x^{-1} \in H$. Hypothesis with $a = e$ and $b = x$ i.e H is closed under taking inverses.

  Closure: Let $x, y \in H$. We've seen $y^{-1} \in H$. Thus $x(y^{-1})^{-1} = xy \in H$, and H is closed. ☆

  $a = x, b = y^{-1}$

  Note:

  We know $(y^{-1})^{-1} = y$ because inverses are unique: $y\cdot y^{-1} = y^{-1}\cdot y = e \Rightarrow (y^{-1})^{-1}\cdot y^{-1} = y^{-1}\cdot(y^{-1})^{-1} = e$

  - Other direction left as an exercise: If $H \leq G$, then $H \neq \emptyset$ and $ab^{-1} \in H$ whenever $a, b \in H$ □

- <u>Two-Step Subgroup Test</u> Let G be a group and H a <u>nonempty</u> subset. Then H is a subgroup if and only if the following two conditions are satisfied:
  1) H is closed under the group operation of G
  2) H is closed under taking inverses

Proof

Suppose H satisfies the hypotheses of the two-step test. We will apply the one-step test.

Nonempty: $H \neq \emptyset$ by hypothesis.

Let $a, b \in H$. Since H is closed under taking inverses, $b^{-1} \in H$. Since H is closed under the group operation, $ab^{-1} \in H$. We just showed that H satisfies the hypotheses of the one-step test. Thus, by the one-step test, $H \leq G$. $\square$

- **Def** (Overloading Notation) for a in a group G, define $\langle a \rangle = \{a^n : n \in \mathbb{Z}\}$. We will end up calling this the cyclic subgroup generated by a, but we don't know that yet
  - Claim: $\langle a \rangle \leq G$
    - **Proof** We apply the one-step subgroup test.

      We see $a^1 = a \in \langle a \rangle$, so $\langle a \rangle \neq \emptyset$.

      Let $x, y \in \langle a \rangle$. By def. of $\langle a \rangle$, $x = a^k$ and $y = a^m$ for some $k, m \in \mathbb{Z}$.

      We see $y^{-1} = a^{-m} \in \langle a \rangle$ since $m \in \mathbb{Z}$, so $-m \in \mathbb{Z}$. ← this kind of proves by 2-step test we can remove this for 1-step

      Now we compute $xy^{-1} = a^k a^{-m} = a^{k-m} \in \langle a \rangle$ since $k - m \in \mathbb{Z}$.

      The one-step test lets us conclude $\langle a \rangle \leq G$. $\square$

- Claim: $\langle a \rangle =$ the smallest subgroup containing $a = \{a^n : n \in \mathbb{Z}\}$ and $\leq$

  In terms of containment: If K is a subgroup containing a, then $\langle a \rangle \subseteq K$

  Proof Let H be the smallest subgroup containing a. $\subseteq$ Since $a \in \langle a \rangle$, we see $H \subseteq \langle a \rangle$.

  $\supseteq$ Conversely, let $x \in \langle a \rangle$. By def., $x = a^m$ for some $m \in \mathbb{Z}$. If $m = 0$, $a^0 = e \in H$, so $x \in H$ since subgroups contain identity. If $m > 0$, then $x = a^m = \underbrace{a \cdot \ldots \cdot a}_{m \text{ times}} \in H$ due to closure of a subgroup. If $m < 0$, then $x = \underbrace{a^{-1} \cdot \ldots \cdot a^{-1}}_{m \text{ times}} \in H$ since subgroups are closed & contain inverses. We've shown containment both ways, so $H = \langle a \rangle$. $\square$

Ex Consider $4 \in (\mathbb{Z}, +)$

$\langle 4 \rangle = 4\mathbb{Z} = \{4n : n \in \mathbb{Z}\}$ ← integer multiples, $\underbrace{4 \cdot \ldots \cdot 4}_{n \text{ times}}$

$\langle 2 \rangle$ is a subgroup

Remarks on notation:

- We will use $\subset$ and $\subseteq$ interchangeably for containment generally
- We will use $\subsetneq$ to specify proper containment
- Various notations for the set of elements of $\mathbb{Z}$ mod m
  - $U(m)$
  - $\mathbb{Z}_m$
  - $\mathbb{Z}/m\mathbb{Z}$
  - $\mathbb{Z}/m\mathbb{Z}^\times$
  - $\mathbb{Z}/m\mathbb{Z}^*$
  - $\mathbb{Z}_{15}^\times$
  - $\mathbb{Z}_{15}^*$

  specify multiplicative